NSC GRADE 12 · MATHEMATICS

Paper 2: every-way-it's-asked mock

One mock paper in the real NSC question order. Under each question number you get every way that question was asked in the November papers 2016–2025.

How to use it

- The real paper is 150 marks in 3 hours: statistics (about 20), analytical geometry (about 40), trigonometry (about 50) and Euclidean geometry (about 40). This mock keeps that order. The real papers number the questions slightly differently each year (10 or 11 questions); the topics always come in this order.
- Each question here holds ALL the variants, so it is much longer than in the real paper. Do one question at a time; the real exam picks a handful of these.
- Under every sub-question, **Asked:** years shows which November papers asked it that way. *Exam Guidelines* means it is in the official DBE scope but was not asked in these ten papers.
- Use the official information sheet (file 4) as in the exam. Give reasons for every statement in Euclidean geometry. Round off to TWO decimal places unless told otherwise. Diagrams are not necessarily drawn to scale.
- Scenarios and numbers are new (not copied), built on the real question patterns. Every memo number was calculated, not guessed.

Q	Topic	Marks here
1	Data Handling: Two Variables (Regression)	30
2	Data Handling: One Variable	70
3	Analytical Geometry: Straight Lines	63
4	Analytical Geometry: The Circle	59
5	Trigonometry: Values, Reduction, Identities And Equations	175
6	Trigonometry: Graphs	67
7	Trigonometry: 2D And 3D Problems	51
8	Euclidean Geometry: Circles (Calculations And Theorem Proofs)	74
9	Euclidean Geometry: Circle Riders (Prove It)	70
10	Euclidean Geometry: Proportionality And Similarity	91
	Total in this mock	**750**

The official past papers and memos for 2016–2025 are in the folder next to this file, if you want to see the originals.

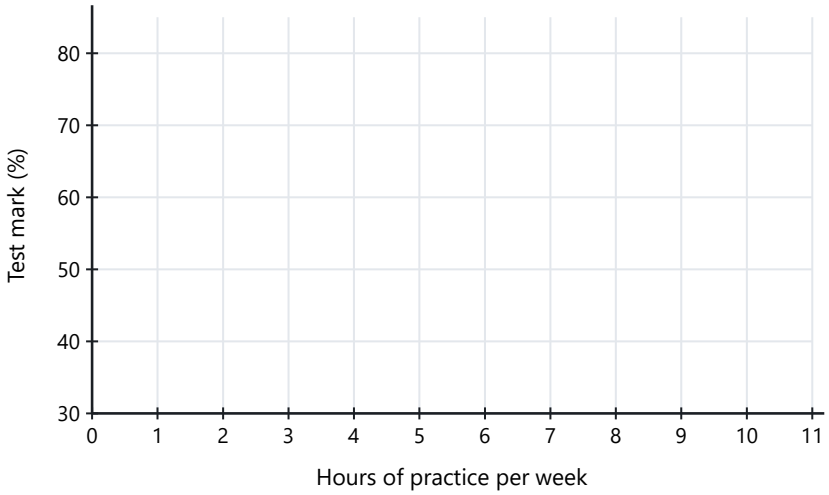
QUESTION 1: DATA HANDLING: TWO VARIABLES (REGRESSION) [30 marks of variants]

Every year: a table and/or scatter plot of two variables. Always: the least squares regression line and a prediction. Usually: the correlation coefficient and what it means. Often a twist: mean and SD of one variable, a point far from the line, or what a or b means in context. 5–12 marks.

A teacher recorded how many hours per week 12 learners spent practising Mathematics and the mark (%) each learner got in a test.

Hours of practice per week (x)	1	2	2	3	4	5	5	6	7	8	9	10
Test mark in % (y)	35	40	46	44	52	50	58	61	60	70	72	79

1.1 On the grid provided, draw a scatter plot to represent the data. (3)



Asked: 2021

1.2 Use the scatter plot to comment on the strength of the relationship between the hours of practice and the test mark. (1)

Asked: 2016, 2018, 2020 (trend)

1.3 Determine the equation of the least squares regression line for the data. (3)

Asked: 2016, 2017 (values of a and b), 2019, 2020, 2021, 2022, 2023, 2024, 2025

1.4 Write down the correlation coefficient for the data. (1)

Asked: 2016, 2019, 2020, 2021, 2023, 2024, 2025

1.5 Interpret your answer to QUESTION 1.4. (1)

Asked: 2020, 2023

1.6 Predict the test mark of a learner who practises 6,5 hours per week. (2)

Asked: All ten years

1.7 Use the correlation coefficient to comment on the reliability of your prediction in QUESTION 1.6. (2)

Asked: 2021, 2022, 2025

1.8 Draw the least squares regression line on the scatter plot. (2)

Asked: 2016, 2020

1.9 Use your answer to QUESTION 1.3 to write down the average increase in test mark for each extra hour of practice per week. (1)

Asked: 2025

1.10 Another learner practised 11 hours per week and got 48%. If this point is added to the data, will the gradient of the regression line increase or decrease? Motivate your answer without further calculation. (2)

Asked: 2017

1.11 A learner who practises 9 hours per week is told that he will score 30%. Which of the following is the most likely response? Give the letter only. (2)

A: Agree, because there is a very strong positive correlation.
B: Agree, because 30% lies on the regression line.
C: Disagree, because the point (9 ; 30) lies very far from the regression line.
D: Disagree, because the correlation is negative.

Asked: 2019

1.12 Calculate the mean and the standard deviation of the test marks. (3)

Asked: 2022, 2023, 2024 (mean only)

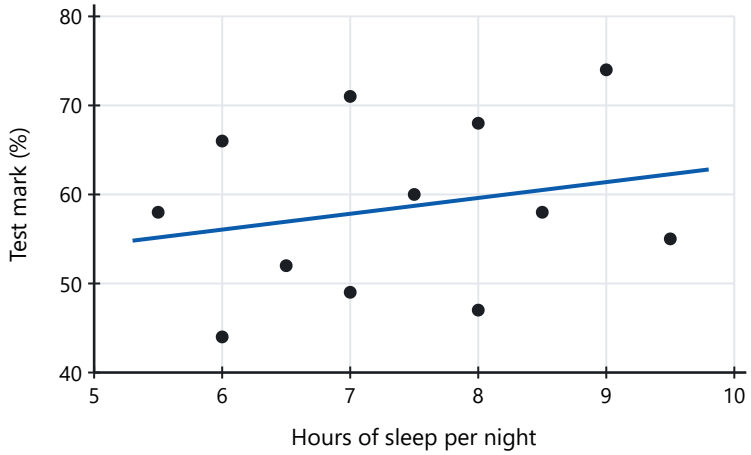
1.13 How many learners scored less than one standard deviation below the mean? (2)

Asked: 2022, 2023

1.14 The teacher uses the regression line as the minimum target for each learner. Determine the largest increase in marks that any ONE learner needs to reach the target. (2)

Asked: 2024

The same learners' hours of sleep per night and test marks are shown in the scatter plot below, with the least squares regression line drawn.



1.15 Which ONE of these correlation coefficients best fits the data? A: $r = 0{,}91$ B: $r = -0{,}62$ C: $r = 0{,}21$ (1)

Asked: 2018

1.16 Use the scatter plot and the regression line to motivate your answer to QUESTION 1.15. (1)

Asked: 2018, 2022 (why not a good indicator)

1.17 In another study the regression line linking SHOE SIZE (x) to HEIGHT in cm (y) of teenagers is $\hat{y} = 82{,}4 + 3{,}9x$. Explain why the point $(0; 82{,}4)$ has no meaning in this context. (1)

Asked: 2018

[30]

QUESTION 2: DATA HANDLING: ONE VARIABLE [70 marks of variants]

Every year one set of data: either a list of values (mean, standard deviation, quartiles, box and whisker) or a grouped table/ogive/histogram (cumulative frequency, estimated mean, reading off the ogive). Usually a twist: an unknown in the table, a missing value from the mean, or what happens when every value changes. 8–15 marks.

The waiting times (in minutes) of 15 taxis at a rank on a Monday morning were:
4 ; 6 ; 7 ; 7 ; 8 ; 9 ; 10 ; 10 ; 11 ; 12 ; 13 ; 15 ; 18 ; 22 ; 29

2.1.1 Calculate the mean waiting time. (2)
Asked: 2017, 2020, 2021, 2022, 2023, 2024

2.1.2 Calculate the standard deviation of the waiting times. (1)
Asked: 2017 (given), 2018, 2020, 2021, 2022, 2023, 2025

2.1.3 Determine the number of waiting times that lie WITHIN one standard deviation of the mean. (3)
Asked: 2020, 2025 (outside), 2017, 2021 (above one SD)

2.1.4 Write down the five-number summary of the data. (3)
Asked: 2017 (given for the other group), 2024 (from ogive)

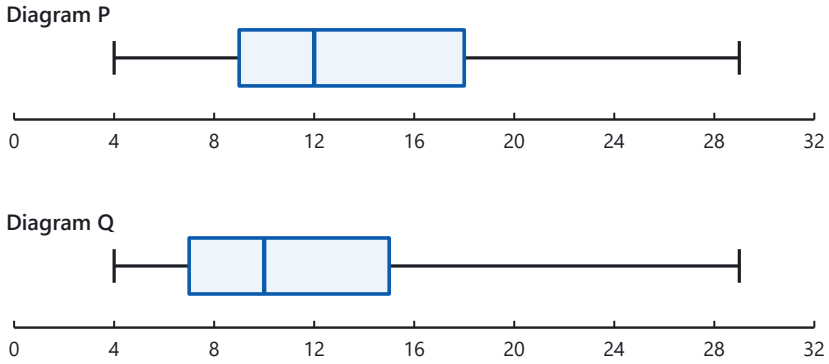
2.1.5 Calculate the interquartile range. (2)
Asked: 2017, 2018, 2024

2.1.6 Calculate the range of the waiting times. (1)
Asked: 2016

2.1.7 Draw a box and whisker diagram to represent the data on the number line provided. (3)
Asked: 2017, 2024

2.1.8 Describe the skewness of the data. (1)
Asked: 2016, 2021, 2025

Below are two box and whisker diagrams. ONE of them represents the waiting times above.



2.1.9 Which ONE of the two diagrams represents the data? Give a reason. (2)
Asked: 2021

2.2.1 At a second taxi rank the five-number summary of the waiting times is $(9; 12; 14; 17; 21)$. Which rank had the lower median waiting time? (1)
Asked: 2017

2.2.2 Three taxis get a prize for the shortest waiting time. How many of the three can come from the second rank? Motivate your answer. (2)
Asked: 2017

2.2.3 Over the same period the first rank's waiting times had the same mean as the times at a third rank, but the third rank's standard deviation was 2,1 minutes. Comment on (a) the total waiting time at each rank, if both ranks had 15 taxis and (b) the variation in waiting times. (2)
Asked: 2018

2.2.4 The stopwatch used at the first rank started 1 minute late every time, so every recorded time is 1 minute too short. How does this influence (a) the mean and (b) the standard deviation? (2)

Asked: 2016, 2024, 2016

2.2.5 Every waiting time at the first rank is increased by 10%. Write down the new mean and standard deviation. (2)

Asked: 2022 (fuel price increase: how the ogive changes)

2.2.6 On the Tuesday the 15 taxis had a mean waiting time of 9 minutes. Calculate how many fewer minutes in TOTAL the taxis waited on Tuesday than on Monday. (2)

Asked: 2021

2.2.7 On Wednesday 9 taxis waited and the MEAN waiting time was 12 minutes. The times of 8 of the taxis were: 3 ; 18 ; 10 ; 21 ; 7 ; 12 ; 14 ; 9. Determine how many of the 9 waiting times were outside ONE standard deviation of the mean. (5)

Asked: 2025

100 learners were asked how many minutes they spent on homework the previous evening. The estimated mean is 60,6 minutes. Two frequencies, a and b, are missing.

Time (minutes)	Number of learners
$0 < x \leq 20$	6
$20 < x \leq 40$	14
$40 < x \leq 60$	a
$60 < x \leq 80$	30
$80 < x \leq 100$	b
$100 < x \leq 120$	5

2.3.1 Show that $a = 28$ and $b = 17$. (5)

Asked: 2019

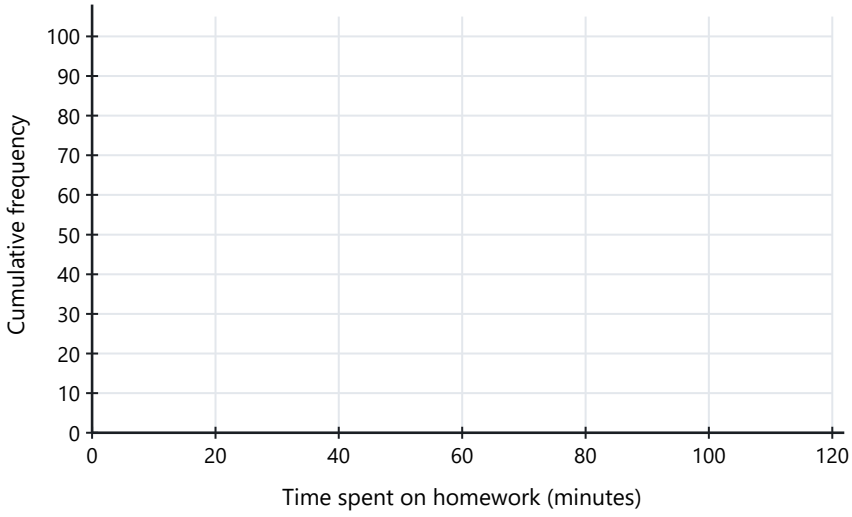
2.3.2 Write down the modal class of the data. (1)

Asked: 2018, 2019, 2022

2.3.3 Complete the cumulative frequency column. (2)

Asked: 2016, 2023, 2019

2.3.4 On the grid provided, draw an ogive (cumulative frequency graph) of the data. (4)



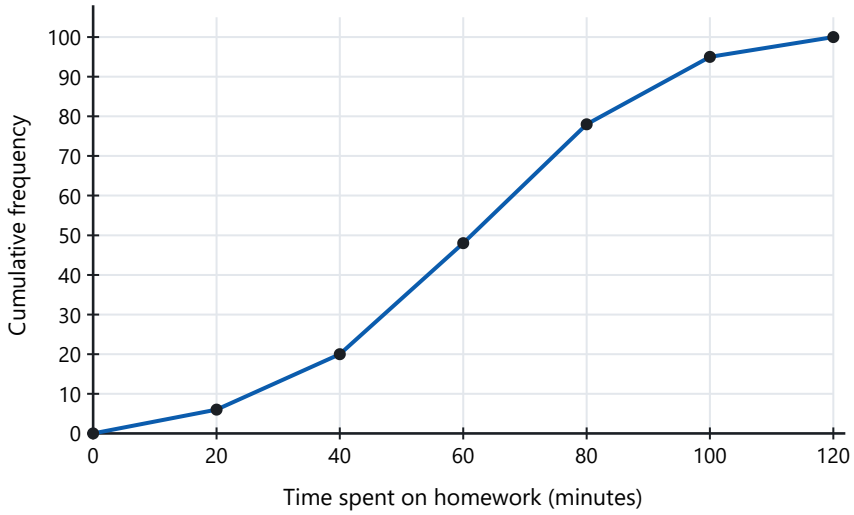
Asked: 2016, 2019

2.3.5 Draw a histogram to represent the data. (3)

Asked: 2025 (from a cumulative table)

The ogive of the data is drawn below.

Time (minutes)	Frequency	Cumulative frequency
$0 < x \leq 20$	6	6
$20 < x \leq 40$	14	20
$40 < x \leq 60$	28	48
$60 < x \leq 80$	30	78
$80 < x \leq 100$	17	95
$100 < x \leq 120$	5	100



2.4.1 Use the ogive to estimate the median time. (1)

Asked: 2024

2.4.2 Use the ogive to write down the total number of learners and the modal class. (2)

Asked: 2018, 2022

2.4.3 Estimate the lower quartile. (1)

Asked: 2018, 2024, 2018 (75th percentile)

2.4.4 Estimate the interquartile range. (2)

Asked: 2018, 2024

2.4.5 How many learners spent more than 70 minutes on homework? (2)

Asked: 2018, 2019, 2022, 2024 (as a percentage)

2.4.6 The shortest and longest times were 5 minutes and 118 minutes. Draw a box and whisker diagram for the data on the number line provided. (2)

Asked: 2024

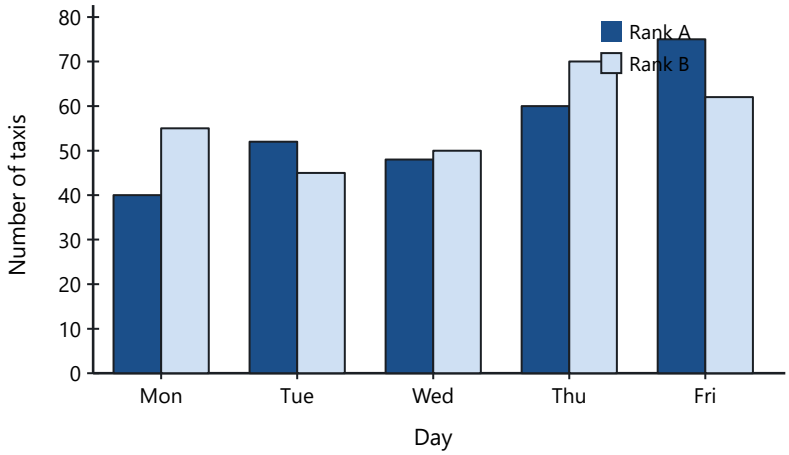
2.4.7 If every learner spent 10 minutes MORE on homework, how would the ogive change? (2)

Asked: 2022

2.4.8 Later it was found that k learners had been absent. Half of them spent $0 < x \leq 20$ minutes and the other half $40 < x \leq 60$ minutes on homework. With them included the estimated mean is 55,5 minutes. Calculate k. (4)

Asked: 2023

The double bar graph shows the number of taxis that left two ranks, A and B, on five days.



2.5.1 On which day was the difference between the two ranks the greatest? (1)

Asked: 2020

2.5.2 Calculate the mean number of taxis per day at each rank. (2)

Asked: 2020

2.5.3 Which ONE of the following statements is correct? Show calculations or reasoning. (2)

A: Rank B had more taxis than rank A on three of the five days.

B: Rank A had the higher mean.

C: The numbers at rank A varied less than at rank B.

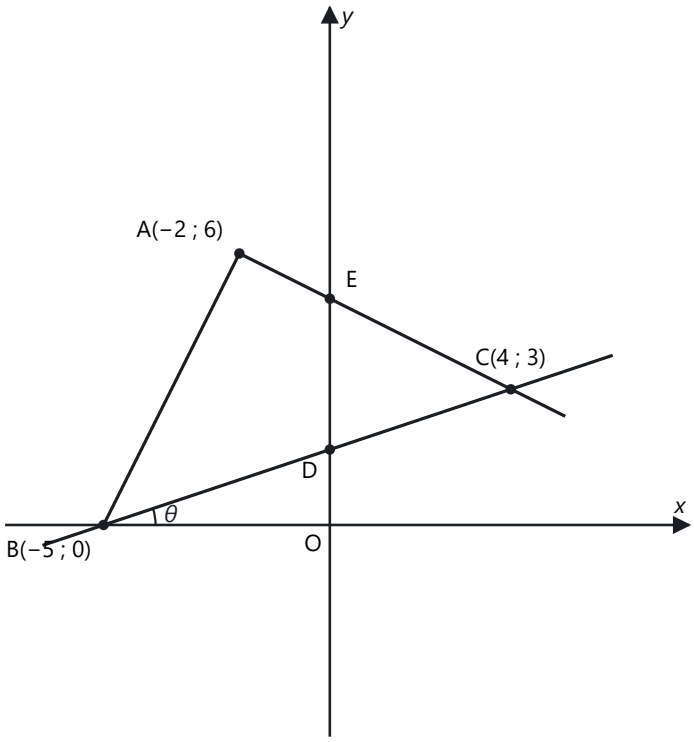
Asked: 2020

[70]

QUESTION 3: ANALYTICAL GEOMETRY: STRAIGHT LINES [63 marks of variants]

Every year: a triangle or quadrilateral on the Cartesian plane. Always: distance, gradient, midpoint, angle of inclination, equation of a line. Then the build-up questions: the angle between two lines, a parallel or perpendicular line, a fourth vertex, an area, an unknown coordinate from a length (quadratic). 17–24 marks.

In the diagram, A(−2 ; 6), B(−5 ; 0) and C(4 ; 3) are the vertices of △ABC. BC cuts the y-axis at D and AC cuts the y-axis at E. The angle of inclination of BC is θ.



3.1 Calculate the length of AB. Leave your answer in simplest surd form. (2)

Asked: 2016, 2018, 2023, 2025

3.2 Calculate the gradient of BC. (2)

Asked: All ten years

3.3 Calculate the size of θ. (2)

Asked: 2016, 2018, 2019, 2022, 2023, 2025

3.4 Calculate the coordinates of M, the midpoint of BC. (2)

Asked: 2016, 2017, 2022

3.5 Determine the equation of BC in the form $y = mx + c$. (3)

Asked: 2017, 2018, 2019, 2021, 2022, 2024, 2025, 2019, 2022 (y-intercept)

3.6 Determine the gradient of the line $3x + 5y = 6$ and show that it passes through M, the midpoint of BC. (3)

Asked: 2020 (5y + 2x + 60 = 0), 2022 (5x − 6y = 8)

3.7 Show that AB ⊥ AC. (2)

Asked: 2020, 2024

3.8 Calculate the size of $\mathrm{A\hat{B}C}$. (4)

Asked: 2018, 2020, 2021, 2022, 2023

3.9 Determine the equation of the line through A parallel to BC. (3)

Asked: 2023, 2022 (CD ∥ BA)

3.10 Calculate the area of △ABC. (3)

Asked: 2018, 2020, 2023, 2025, 2022 (by subtraction)

3.11 Use your answer to 3.8 and the area rule to confirm the area of △ABC. (2)

Asked: 2018

3.12 Calculate the area of quadrilateral ABDE. (4)

Asked: 2019, 2020, 2021, 2022

3.13 Calculate the coordinates of the point that is equidistant from A, B and C. (3)

Asked: 2023

3.14 Prove that NM ∥ AC, where N is the midpoint of AB. (4)

Asked: 2017

3.15 ABKC, in that order, is a square. Determine the coordinates of K. (3)

Asked: 2017, 2018, 2019, 2021, 2025 (parallelogram)

3.16 F is a point on AC, between A and C, such that BF = $\sqrt{65}$. Calculate the coordinates of F. (5)

Asked: 2016, 2018, 2021, 2024

3.17 G(p ; 6) is a point such that CG ⊥ BC. Write down the gradient of CG in terms of p, then calculate p. (2)

Asked: 2016, 2021, 2024 (unknown on a line)

3.18 T is the point on BC such that AT ⊥ BC. Calculate the coordinates of T and hence the length AT. (5)

Asked: 2025

3.19 ABCP, in that order, is a parallelogram. Use your answer to 3.18 to calculate its area. (3)

Asked: 2020, 2025

3.20 G is a point on AB and H is a point on AC such that GH ∥ BC and AG : GB = 1 : 2. Calculate $\frac{\text{Area } \triangle AGH}{\text{Area } \triangle ABC}$. (3)

Asked: 2024

3.21 △ABC is reflected about the y-axis to form △A′B′C′. Calculate the size of $\text{C}\hat{\text{E}}\text{C}'$. (3)

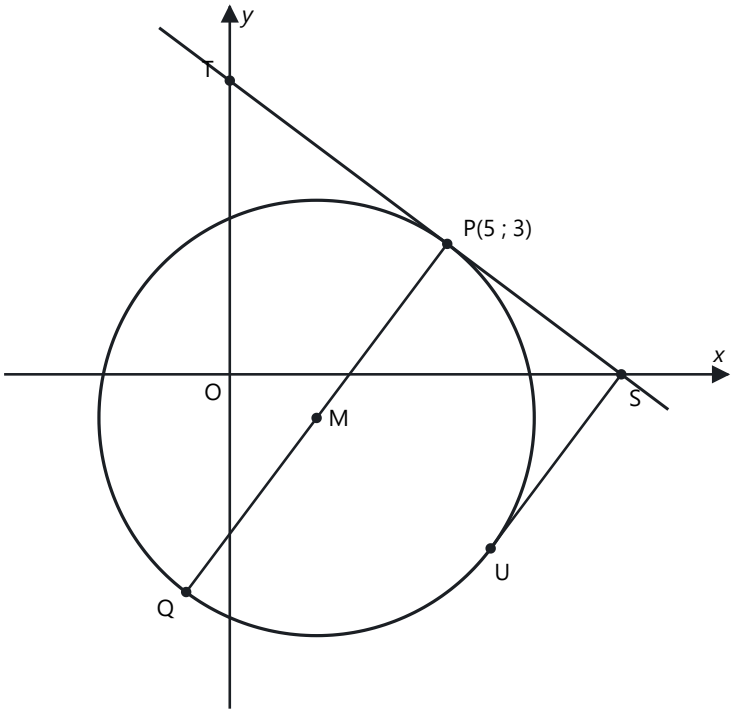
Asked: 2019, 2016 (angle at the y-axis)

[63]

QUESTION 4: ANALYTICAL GEOMETRY: THE CIRCLE [59 marks of variants]

Every year: one or two circles with a tangent. Always: centre and radius, the equation of the circle, the equation of a tangent (radius ⊥ tangent). Then: an unknown point on the circle, values of k for a line to cut/touch/miss the circle, length of a tangent from an outside point, two circles touching, a translation or reflection. 15–23 marks.

In the diagram, M is the centre of the circle $x^2 + y^2 - 4x + 2y - 20 = 0$. P(5 ; 3) and Q are points on the circle such that PQ is a diameter. The tangent to the circle at P cuts the x-axis at S and the y-axis at T. SU is a second tangent from S, touching the circle at U.



4.1 Determine the coordinates of M and the length of the radius. (3)

Asked: 2018 (write down F), 2024 (show general form), 2023

4.2 Show that the equation of the circle can be written as $x^2 + y^2 - 4x + 2y - 20 = 0$ if you are given only the centre M(2 ; −1) and the point P(5 ; 3) on it. (3)

Asked: 2017, 2020, 2022, 2024

4.3 Calculate the coordinates of Q. (2)

Asked: 2016, 2019, 2022, 2016 (circle on a diameter)

4.4 Determine the equation of the tangent ST in the form $y = mx + c$. (4)

Asked: 2016, 2017, 2019, 2020, 2021, 2022, 2023, 2024, 2025

4.5 Show that the line $x = 7$ is a tangent to the circle and write down the point of contact. (3)

Asked: 2016 (motivate whether a line is a tangent), 2018 (tangent x = 22), 2020

4.6 For which values of k will the line $y = -\frac{3}{4}x + k$ be a secant (cut the circle at two points)? (3)

Asked: 2017, 2022, 2019 (not touch: k outside the interval)

4.7 Show that S lies outside the circle and calculate the length of the tangent SP. (3)

Asked: 2018, 2022

4.8 Give a reason why SP = SU. (1)

Asked: 2018, 2021

4.9 Calculate the size of $\mathrm{P\hat{S}U}$. (4)

Asked: 2025

4.10 Calculate the area of quadrilateral MPSU. (2)

Asked: 2017 (area of △SMK), 2019

4.11 R(d ; 3) is another point on the circle, with $d < 0$. Calculate d. (2)

Asked: 2023, 2024

4.12 Calculate the coordinates of the points where the line $y = 2x - 10$ cuts the circle. (4)

Asked: 2021 (second intersection point), 2025 (x-intercepts)

4.13 Point A(t ; $t + 8$) lies outside the circle. AB is a tangent to the circle at B. Show that $AB = \sqrt{2t^2 + 14t + 60}$. (2)

Asked: 2022

4.14 Hence determine the minimum length of AB. (3)

Asked: 2022

4.15 Another circle with centre N(2 ; 11) and radius p touches the circle with centre M. Determine the possible values of p. (4)

Asked: 2018, 2020

4.16 The circle $(x + 10)^2 + (y + 1)^2 = 4$ is translated k units to the right (along the line $y = -1$). Calculate the values of k for which it touches the circle with centre M INTERNALLY. (4)

Asked: 2023

4.17 The circle with centre M is translated 8 units to the left. Write down the minimum distance between the new circle and the y-axis. (1)

Asked: 2025

4.18 The circle with centre M is reflected about the line $y = x$. Calculate the distance between the two centres. (3)

Asked: 2021

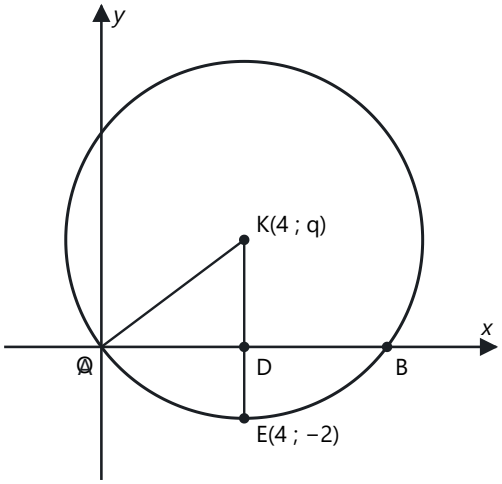
4.19 The circle and its image intersect at two points. Write down the midpoint of their common chord. (2)

Asked: 2021

4.20 A circle with centre K(−4 ; 3) touches the x-axis. Write down its equation. (2)

Asked: 2019 (circle touching both axes), 2025

A circle with centre K(4 ; q), $q > 0$, cuts the x-axis at A and B. D is the midpoint of AB, KD ⊥ AB and AD = $q + 1$. KD produced meets the circle at E(4 ; −2).



4.21 Show that $q = 3$. (4)

Asked: 2025

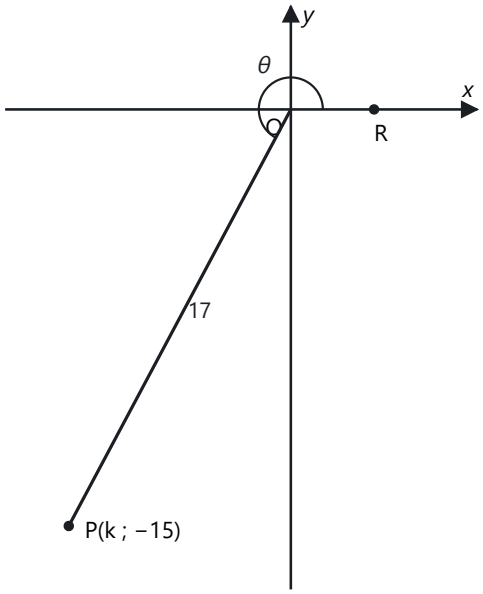
[59]

QUESTION 5: TRIGONOMETRY: VALUES, REDUCTION, IDENTITIES AND EQUATIONS

[175 marks of variants]

Every year 20–30 marks of algebraic trig: a point on the Cartesian plane or 'given sin A = p' (answer WITHOUT a calculator), reduction formulae to simplify, a compound-angle derivation, an identity to prove, and a general solution. The pieces are mixed differently every year.

In the diagram, P(k ; −15) is a point in the third quadrant with OP = 17. R is a point on the positive x-axis and the reflex angle R$\hat{O}$P $= \theta$. Answer 5.1.1 to 5.1.5 WITHOUT using a calculator.



5.1.1 Calculate the value of k. (2)

Asked: 2017, 2018

5.1.2 Write down the value of $\tan\theta$. (1)

Asked: 2017, 2018, 2020

5.1.3 Determine the value of $\cos(180^\circ + \theta)$. (2)

Asked: 2018, 2022, 2016

5.1.4 Determine the value of $\sin 2\theta$. (3)

Asked: 2023, 2017 (cos 2β), 2024 (cos 2A), 2017, 2024 (cos 2θ)

5.1.5 Determine the value of $\sin(\theta + 60^\circ)$ in the form $\dfrac{a + b\sqrt{3}}{34}$. (4)

Asked: 2018

5.1.6 If it is further given that $\sin\beta = \dfrac{4}{5}$ and $90^\circ < \beta < 360^\circ$, calculate $\cos(\theta - \beta)$ without a calculator. (4)

Asked: 2024

5.1.7 S(a ; b) is a point such that R$\hat{O}$S $= \theta - 90^\circ$ and OS = 34. Calculate b without a calculator. (4)

Asked: 2020

5.1.8 Use a calculator to calculate $\tan(2\theta - 40^\circ)$ correct to ONE decimal place. (3)

Asked: 2018

It is given that $\cos 32^\circ = t$. Answer 5.2.1 to 5.2.4 WITHOUT using a calculator, in terms of t.

5.2.1 $\sin 32^\circ$ (2)

Asked: 2016, 2021, 2025, 2021 (tan)

5.2.2 $\cos 64^\circ$ (2)

Asked: 2021 (cos 108°)

5.2.3 $2\sin^2 74^\circ$ (3)

Asked: 2019

5.2.4 $\sin 2^\circ$ (4)

Asked: 2025 (sin 10° from tan 50°)

5.2.5 If instead $\tan 35^\circ = k$, express $\cos 55^\circ$ in terms of k. (2)

Asked: 2025

5.3 Given $5\cos x + 3 = 0$ and $x \in (180^\circ; 360^\circ)$. Without a calculator, determine $\tan x$ and $\sin(360^\circ + x)$. (3)

Asked: 2022, 2023 (sin β = 1/3 with β ∈ (90° ; 270°))

Given $\cos 2\beta = \frac{7}{25}$ and $0^\circ \leq \beta \leq 90^\circ$. Determine, without a calculator:

5.4.1 $\cos \beta$ (3)

Asked: 2016

5.4.2 $\sin \beta$ (2)

Asked: 2016

5.4.3 $\cos(\beta + 45^\circ)$ (3)

Asked: 2016

5.5.1 Simplify to ONE trigonometric ratio: $\frac{\sin(180^\circ + x) \cdot \cos(90^\circ - x)}{\cos(90^\circ + x) \cdot \tan(-x)}$ (6)

Asked: 2016, 2017, 2019, 2021, 2022, 2025

5.5.2 Simplify $\frac{\sqrt{1 - \cos^2 2A}}{\cos(-A) \cdot \cos(90^\circ + A)}$ completely, given $0^\circ < A < 90^\circ$. (5)

Asked: 2016

5.5.3 Determine the value of $\frac{\sin 2x \cdot \cos(-x) + \cos 2x \cdot \sin(360^\circ - x)}{\sin(180^\circ + x)}$. (5)

Asked: 2020

5.5.4 Without using a calculator, determine the value of $\frac{\tan 150^\circ \cdot \sin 300^\circ}{\cos 210^\circ \cdot \sin 60^\circ}$. (5)

Asked: 2020, 2021, 2022

5.5.5 Without using a calculator, determine the value of $\frac{4\sin 15^\circ \cos 15^\circ}{\cos^2 75^\circ - \sin^2 75^\circ}$. (4)

Asked: 2019

5.5.6 Evaluate without a calculator: $\cos^2 20^\circ + \cos^2 21^\circ + \cos^2 22^\circ + \ldots + \cos^2 70^\circ$. (5)

Asked: 2018

5.5.7 Given $x - \frac{1}{x} = 2\sin A$ and $x^2 + \frac{1}{x^2} = 3$. Determine the value of $\cos 2A$ without a calculator. (5)

Asked: 2020

5.5.8 If $\sin \alpha = p$ and α is acute, express $4\sin\left(\frac{\alpha}{2} + 45^\circ\right)\cos\left(\frac{\alpha}{2} + 45^\circ\right)$ in terms of p. (4)

Asked: 2024

5.6.1 Use $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ to derive a formula for $\cos(\alpha + \beta)$. (3)

Asked: 2021, 2024

5.6.2 Use $\cos(\alpha - \beta)$ to derive a formula for $\sin(\alpha + \beta)$. (3)

Asked: 2016, 2023 (sin(α − β))

5.6.3 Prove that $\sin(A+B)-\sin(A-B)=2\cos A\sin B$ and hence, without a calculator, show that $\sin 75^\circ-\sin 15^\circ=\frac{\sqrt{2}}{2}$. (4)

Asked: 2017

5.7.1 Prove the identity: $\frac{\sin 2x+\sin x}{\cos 2x+\cos x+1}=\tan x$ (5)

Asked: 2018, 2021, 2023, 2025

5.7.2 For which values of x in $[0^\circ;360^\circ]$ is the identity in 5.7.1 undefined? (3)

Asked: 2023

5.7.3 Prove that $\cos(x+y)\cos(x-y)=\cos^2 x-\sin^2 y$. (4)

Asked: 2022

5.7.4 Hence, without a calculator, determine the value of $\cos^2 52{,}5^\circ-\sin^2 7{,}5^\circ$. (3)

Asked: 2022

5.7.5 Prove that $\frac{\cos^4 x-\sin^4 x}{\cos x-\sin x}=\cos x+\sin x$. (4)

Asked: 2023 ($\cos^4x + \sin^2x\cos^2x$), 2025

5.7.6 Prove that $\frac{\cos x+\sin x}{\cos x-\sin x}-\frac{\cos x-\sin x}{\cos x+\sin x}=2\tan 2x$. (5)

Asked: 2018

5.7.7 Prove that $\frac{\cos^3 x-\sin^3 x}{\cos x-\sin x}=1+\sin x\cos x$. (3)

Asked: 2025

5.7.8 Simplify fully: $\frac{2\cos^2 x+\cos x-1}{\cos(180^\circ+x)-1}$ (5)

Asked: 2021

5.7.9 Simplify $\frac{\sin 3x-\sin x}{2\cos 2x}$ to a single trigonometric ratio. (6)

Asked: 2023 (sin 3x + sin x), 2021 (2cos 6x cos 4x − cos 10x)

5.7.10 Rewrite $8\sin x\cos^3 x-8\sin^3 x\cos x$ as a single trigonometric ratio, and write down its maximum value and the value of $x\in[0^\circ;90^\circ]$ where it occurs. (5)

Asked: 2022, 2024 (max of g), 2019 (min)

5.7.11 Simplify $\frac{\sin(540^\circ+x)\cos(90^\circ+x)}{\sin(-x)}$ to ONE trigonometric ratio. Hence determine the values of $x\in[0^\circ;360^\circ]$ for which the square root of this expression is real. (5)

Asked: 2025

5.8.1 Determine the values of $x\in[0^\circ;360^\circ]$ for which $6\cos^2 x+5\sin x-7=0$. (6)

Asked: 2020, 2024

5.8.2 Determine the general solution of $(\sin x+3\cos x)(2\cos 2x-1)=0$. (6)

Asked: 2022

5.8.3 Determine the general solution of $\tan x=4\sin x$, where $\cos x>0$. (6)

Asked: 2021

5.8.4 Determine the general solution of $\sin 50^\circ\cos x-\cos 50^\circ\sin x=\cos 2x$. (6)

Asked: 2019, 2023

5.8.5 $\cos^2 x$; $\sin^2 x$; $\sin x\cos x$ are the first three terms of an arithmetic sequence with a constant difference that is NOT zero. Determine the general solution for x. (7)

Asked: 2025

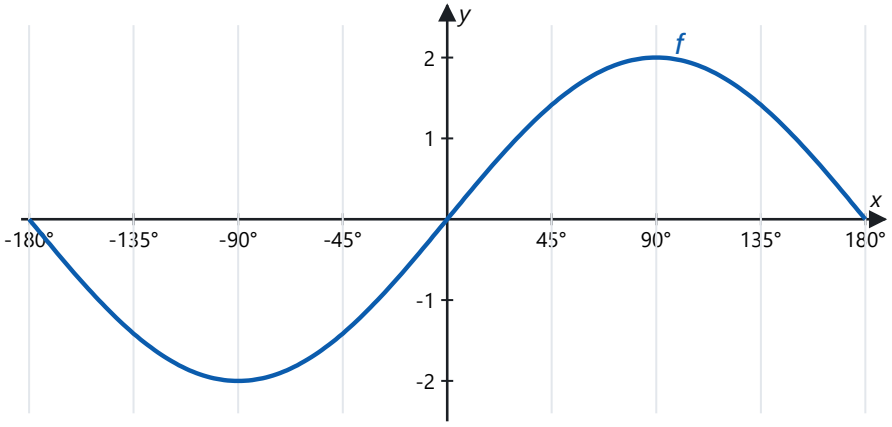
[175]

QUESTION 6: TRIGONOMETRY: GRAPHS

[67 marks of variants]

Every year one or two trig graphs on a restricted interval (often one drawn, one for you to draw). Always: period/amplitude/range and reading off the graphs (where f > g, f·g ≥ 0). Usually: solving f = g algebraically, a translation that gives a new equation. 8–13 marks.

In the diagram, the graph of $f(x) = 2\sin x$ is drawn for $x \in [-180°; 180°]$. Let $g(x) = \cos 2x$.



6.1 On the same system of axes, draw the graph of $g(x) = \cos 2x$ for $x \in [-180°; 180°]$. Show ALL the intercepts with the axes, the turning points and the end points. (4)

Asked: 2016, 2017, 2021, 2024, 2025

6.2 On a new set of axes, draw $q(x) = \cos(x - 60°)$ for $x \in [-180°; 180°]$. Show the intercepts and turning points. (3)

Asked: 2021

6.3 Draw $k(x) = \tan 2x - 1$ for $x \in [-90°; 90°]$. Show the asymptotes, intercepts and end points. (4)

Asked: 2018, 2025

6.4 Write down the period of g and the amplitude of f. (2)

Asked: 2018, 2020, 2022, 2023, 2024, 2025, 2021 (period of f(3x))

6.5 Write down the range of $h(x) = \frac{1}{2}f(x) + 1$. (2)

Asked: 2019, 2021, 2022, 2023, 2025, 2016 (max of f(x) − 3)

6.6 Write down the range of g for $x \in [0°; 60°]$. (2)

Asked: 2023

6.7 A and B are the points of intersection of f and g. Show that the x-coordinates of A and B satisfy $\sin x = \frac{-1 + \sqrt{3}}{2}$. (4)

Asked: 2017

6.8 Hence calculate the coordinates of A and B. (3)

Asked: 2017

6.9 Determine the general solution of $2\sin x - \cos 2x = 0$. (4)

Asked: 2016, 2018, 2024

6.10 Determine the general solution of $2\sin 2x = \cos 2x$. (3)

Asked: 2016

6.11 Use the graphs to determine the values of x for which $f(x) > g(x)$. (2)

Asked: 2016, 2020

6.12 Use the graphs to determine the values of x for which $f(x) \cdot g(x) \geq 0$. (3)

Asked: 2023, 2025

6.13 For which values of $x \in [-180°; 0°]$ is $f(x) + 1 \leq 0$? (2)

Asked: 2023

6.14 For which values of x is f decreasing? (2)

Asked: 2019

6.15 Use the graphs to solve $f(x + 10°) = g(x + 10°)$ for $x \in [-180°; 170°]$. (2)

Asked: 2020, 2022

6.16 The graph of g is translated $45°$ to the LEFT to form the graph of p. Determine the equation of p in its simplest form. (2)

Asked: 2018, 2023, 2025

6.17 D(k ; 1) lies on the graph of $r(x) = -f(x)$. Determine the values of k for $x \in [-180°; 180°]$. (3)

Asked: 2023

6.18 For which value(s) of t will $\cos 2x = t$ have exactly TWO solutions in $[-180°; 180°]$? (2)

Asked: 2022

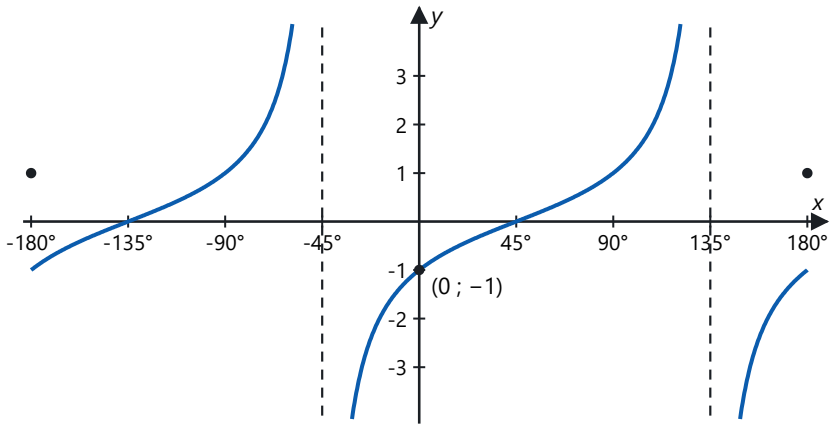
6.19 P and Q are points on g and f respectively, with PQ parallel to the y-axis, for $x \in [-180°; 0°]$. Determine the maximum length of PQ and the value(s) of x where it occurs. (6)

Asked: 2019

6.20 Show that $\sqrt{3}\sin x + \cos x = 2\sin(x + 30°)$ and hence solve $\sqrt{3}\sin x + \cos x \geq 1$ for $x \in [0°; 180°]$. (4)

Asked: 2020

The graph of $h(x) = \tan(x - 45°)$ is drawn below for $x \in [-180°; 180°]$.



6.21 Write down the period of h and the equations of its asymptotes. (2)

Asked: 2018, 2024, 2025

6.22 The point A(t ; 1) lies on h. Determine the general solution of t. (3)

Asked: 2018

6.23 Determine the values of $x \in [-180°; 180°]$ for which $h(x) \geq 1$. (3)

Asked: 2018, 2024 (f(x) ≤ 0)

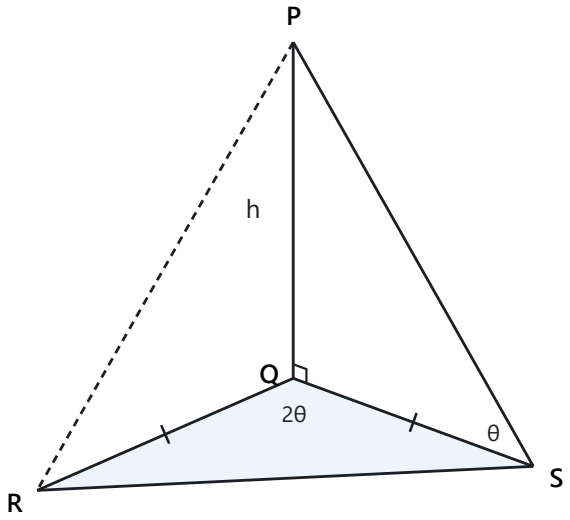
[67]

QUESTION 7: TRIGONOMETRY: 2D AND 3D PROBLEMS

[51 marks of variants]

Every year one problem in 3D: a vertical pole/tower/building and a triangle in the horizontal plane, or a pyramid. Always: one right-angled triangle (tan/sin/cos), then the sine rule, cosine rule or area rule in the other triangle, then a 'show that' formula and a calculation with given values. 6–12 marks.

PQ is a vertical pole of height h metres. Q, R and S lie in the same horizontal plane, with QR = QS. The angle of elevation from S to P is θ and $\mathrm{R\hat{Q}S} = 2\theta$.



7.1.1 Write down the size of $\mathrm{P\hat{Q}S}$. (1)

Asked: 2017

7.1.2 Express QS in terms of h and θ. (2)

Asked: 2017, 2018, 2020, 2023, 2024

7.1.3 Show that $RS = 2h\cos\theta$. (5)

Asked: 2018, 2020 (RS = r√(10 − 6cos 2x)), 2022

7.1.4 If $h = 12$ m and $\theta = 35^\circ$, calculate RS. (2)

Asked: 2020, 2023

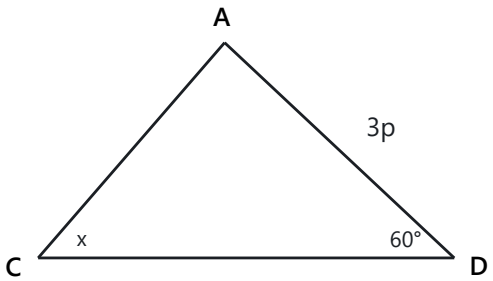
7.1.5 Calculate the area of △QRS if $h = 12$ m and $\theta = 35^\circ$. (3)

Asked: 2019, 2021, 2022

7.1.6 In another design QR = QS = 15 m and the area of △QRS is 100 m². Calculate $\mathrm{R\hat{Q}S}$ (acute). (3)

Asked: 2019, 2023

In △ACD, AD = $3p$, $\mathrm{A\hat{D}C} = 60^\circ$ and $\mathrm{A\hat{C}D} = x$.



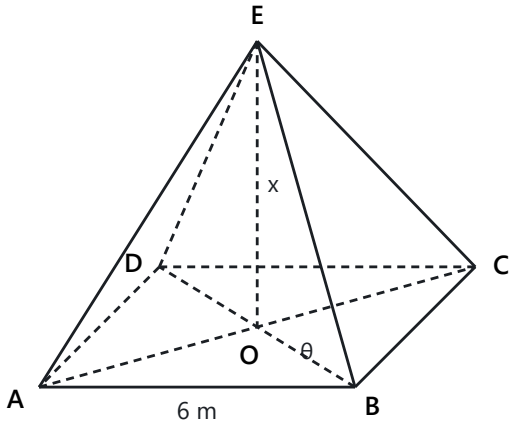
7.2.1 Show that $CD = \dfrac{3p(\sqrt{3}\cos x + \sin x)}{2\sin x}$. (5)

Asked: 2017, 2021, 2022

7.2.2 If $p = 4$ and $x = 50°$, calculate the area of △ACD. (3)

Asked: 2021, 2022

E is the apex of a pyramid with a square base ABCD of side 6 m. O is the centre of the base and EO, the perpendicular height, is x metres. $\mathrm{E\hat{B}A} = \theta$. Volume of a pyramid $= \frac{1}{3} \times$ area of base $\times$ perpendicular height.



7.3.1 Calculate the length of OB. (3)

Asked: 2016

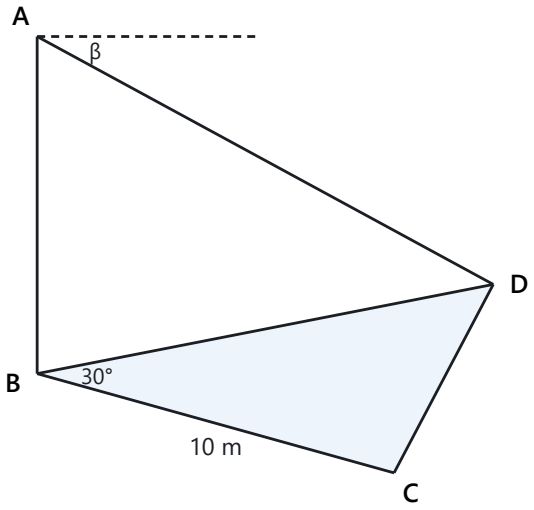
7.3.2 Show that $\cos\theta = \frac{3}{\sqrt{x^2 + 18}}$. (4)

Asked: 2016

7.3.3 If the volume of the pyramid is 48 m³, calculate θ. (4)

Asked: 2016

AB is a vertical cliff. B, C and D lie in the same horizontal plane. The angle of depression from A to D is β. BC = 10 m, $\mathrm{C\hat{B}D} = 30°$ and the area of △BCD is 40 m².



7.4.1 Show that BD = 16 m. (3)

Asked: 2023

7.4.2 If β = 25°, calculate AB. (3)

Asked: 2018, 2023

7.4.3 Calculate CD. (3)

Asked: 2018, 2024, 2025

7.4.4 E is another point in the plane with CE = 12 m and $\hat{CBE} = 30^\circ$ (E on the other side of BC). Calculate BE. (4)

Asked: 2024

7.5 In a horizontal triangle the sides are 7 m, 9 m and 12 m. Calculate the size of the largest angle. (3)

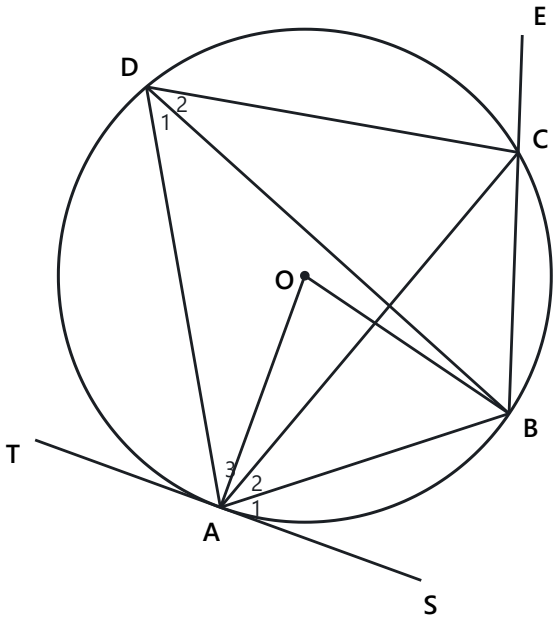
Asked: 2024, 2025 (K̂AC)

[51]

QUESTION 8: EUCLIDEAN GEOMETRY: CIRCLES (CALCULATIONS AND THEOREM PROOFS) [74 marks of variants]

Every year: angles calculated WITH reasons in a circle with a tangent and/or a cyclic quadrilateral, a length from the perpendicular from the centre, and (most years) one theorem proof from the four examinable circle proofs below. 7–17 marks.

In the diagram, O is the centre of the circle and ABCD is a cyclic quadrilateral. TAS is a tangent to the circle at A. BC is produced to E. AC, BD, OA and OB are drawn. $\hat{A}_1 = 38^\circ$, $\hat{A}_3 = 50^\circ$ and $\text{A}\hat{\text{D}}\text{C} = 70^\circ$. Calculate, with reasons, the size of:



8.1.1 $\hat{D}_1$ (2)

Asked: 2016, 2018, 2020, 2024, 2025

8.1.2 $\text{A}\hat{\text{O}}\text{B}$ (2)

Asked: 2020, 2022, 2025

8.1.3 $\text{O}\hat{\text{A}}\text{B}$ (3)

Asked: 2018, 2020, 2025

8.1.4 $\hat{A}_2$ (3)

Asked: 2017, 2018, 2019

8.1.5 $\text{C}\hat{\text{B}}\text{D}$ (2)

Asked: 2016, 2017, 2019

8.1.6 $\text{A}\hat{\text{B}}\text{C}$ (2)

Asked: 2019, 2021, 2022, 2024

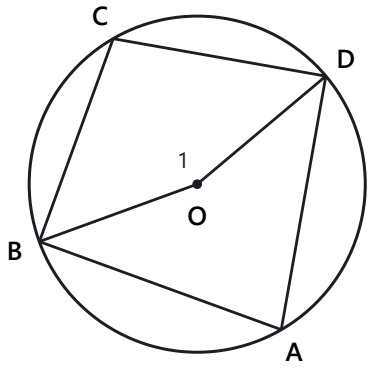
8.1.7 $\text{D}\hat{\text{C}}\text{E}$ (2)

Asked: 2019, 2021, 2024

8.1.8 Is TS parallel to BD? Motivate your answer. (3)

Asked: 2018 (∥ lines), 2025 (prove PQ ∥ SR)

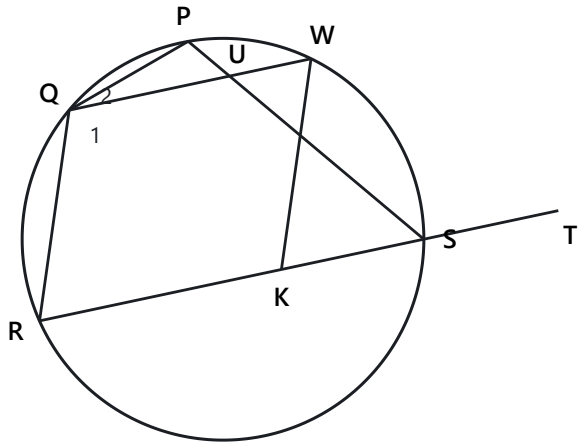
In the diagram, O is the centre of the circle and ABCD is a cyclic quadrilateral. OB and OD are drawn. $\hat{O}_1 = 3x + 40^\circ$ and $\hat{C} = 2x + 20^\circ$.



8.2 Calculate, with reasons, the value of x. (5)

Asked: 2023

PQRS is a cyclic quadrilateral. RS is produced to T. K is a point on RS and W a point on the circle such that QRKW is a parallelogram. PS and QW intersect at U. $\hat{PST} = 128^\circ$ and $\hat{Q}_1 = 110^\circ$.



8.3.1 Calculate, with reasons, $\hat{R}$. (2)

Asked: 2019

8.3.2 Calculate, with reasons, $\hat{P}$. (2)

Asked: 2019

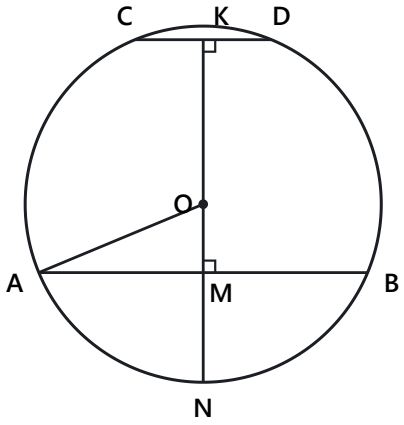
8.3.3 Calculate, with reasons, $\hat{PQW}$. (3)

Asked: 2019

8.3.4 Calculate, with reasons, $\hat{PUQ}$. (2)

Asked: 2019

In the diagram, O is the centre of the circle. AB = 24 units and OM ⊥ AB. OM produced meets the circle at N. CD = 10 units, OK ⊥ CD and CD ∥ AB, on the other side of O.



8.4.1 Give a reason why AM = 12 units. (1)

Asked: 2016, 2021, 2023

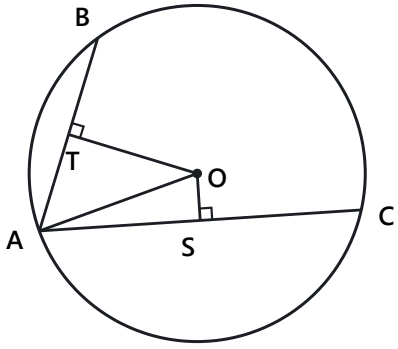
8.4.2 If OM = 5 units, calculate the radius of the circle. (3)

Asked: 2023

8.4.3 Calculate the distance between the two chords. (4)

Asked: 2016, 2021

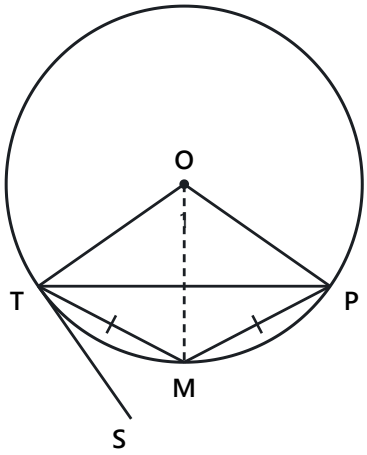
In another circle with centre O, chords AB = 30 and AC = 48 are drawn. OT ⊥ AB and OS ⊥ AC, and $OS = \frac{7}{20}OT$.



8.5 Calculate the radius of the circle. (5)

Asked: 2016

O is the centre of the circle. ST is a tangent to the circle at T. M and P are points on the circle such that TM = MP. OT, OP and OM are drawn. Let $\hat{O}_1 = x$.



8.6.1 Give a reason why $\mathrm{T\hat{O}M}$ = $\mathrm{M\hat{O}P}$. (1)

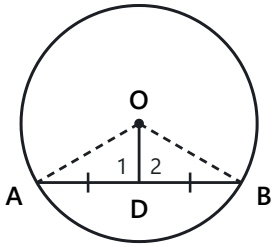
Asked: 2019 (reason list)

8.6.2 Prove that $\mathrm{S\hat{T}M} = \frac{1}{4}x$. (7)

Asked: 2019

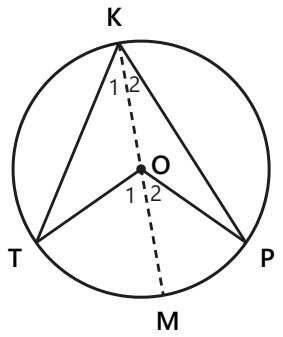
THEOREM PROOFS. Only these circle proofs are examinable. Learn the construction and every reason word for word; it is 5 free marks when one comes up.

8.7.1 In the diagram, O is the centre of the circle and OD bisects chord AB. Prove the theorem that states that the line from the centre of a circle that bisects a chord is perpendicular to the chord, i.e. OD ⊥ AB. (5)



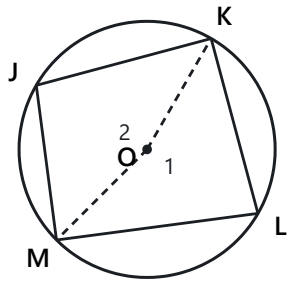
Asked: 2022

8.7.2 In the diagram, O is the centre of the circle and K, T and P are points on the circle. Prove the theorem that states that the angle subtended by an arc at the centre of a circle is double the size of the angle subtended by the same arc at the circle, i.e. $\mathrm{T\hat{O}P} = 2\mathrm{T\hat{K}P}$. (5)



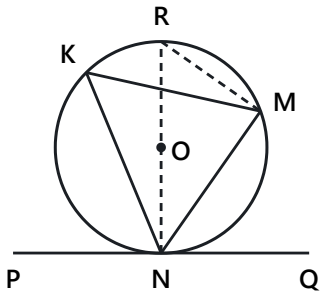
Asked: 2023, 2025

8.7.3 In the diagram, O is the centre of the circle and JKLM is a cyclic quadrilateral. Prove the theorem that states that the opposite angles of a cyclic quadrilateral are supplementary, i.e. $\hat{J} + \hat{L} = 180^\circ$. (5)



Asked: 2018

8.7.4 In the diagram, chords KM, MN and KN are drawn and PNQ is a tangent to the circle at N. Prove the theorem that states that the angle between a tangent and a chord is equal to the angle in the alternate segment, i.e. $\mathrm{M\hat{N}Q} = \hat{K}$. (5)



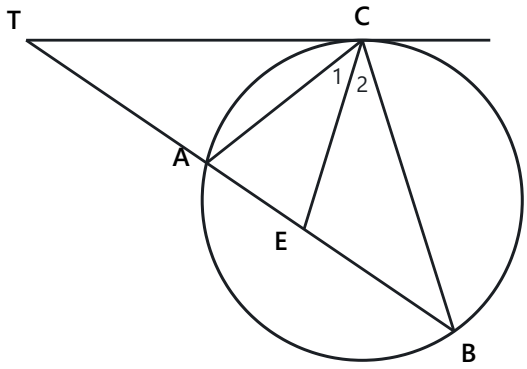
Asked: 2017, 2020, 2021

[74]

QUESTION 9: EUCLIDEAN GEOMETRY: CIRCLE RIDERS (PROVE IT) [70 marks of variants]

Every year 10–25 marks of 'prove that' in a circle diagram: name angles equal to x, prove a cyclic quadrilateral (or that it is NOT one), prove a line is a tangent (converse tan chord), prove sides equal, lines parallel, or a chord is a diameter. Every step needs a reason.

In the diagram, TC is a tangent to the circle at C. TAB is a straight line with A and B on the circle. CE bisects $\mathrm{A\hat{C}B}$, with E on AB. Let $\hat{B} = x$ and $\hat{C}_1 = \hat{C}_2 = y$.



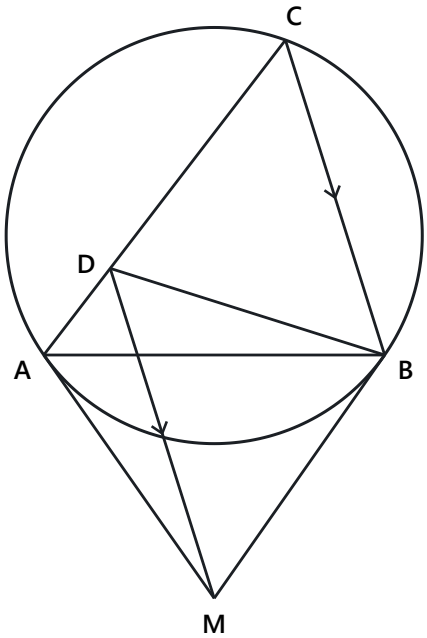
9.1.1 Give a reason why $\mathrm{T\hat{C}A} = x$. (1)

Asked: 2016, 2018, 2019, 2021, 2024, 2025

9.1.2 Prove that TC = TE. (5)

Asked: 2019 (prove △ isosceles), 2023 (AD = AE type)

In the diagram, MA and MB are tangents to the circle at A and B. C is a point on the circle and D is a point on chord AC such that DM ∥ CB. DB and AB are drawn. Let $\mathrm{M\hat{A}B} = x$.



9.2.1 Name, with reasons, THREE other angles each equal to x. (6)

Asked: 2016, 2019, 2024

9.2.2 Prove that ADBM is a cyclic quadrilateral. (3)

Asked: 2016, 2019, 2021, 2022, 2023

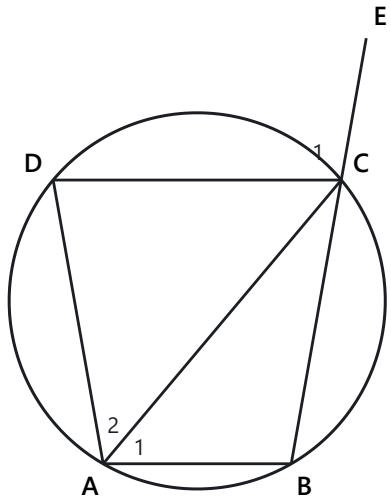
9.2.3 Prove that DM bisects $\mathrm{A\hat{D}B}$. (3)

Asked: 2020 (KN bisects $\mathrm{O\hat{K}P}$)

9.2.4 Prove that DB = DC. (4)

Asked: 2019 (△WRV isosceles), 2024 (AD = DC)

In the diagram, ABCD is a cyclic quadrilateral and BC is produced to E. AC is drawn. $\hat{A}_1 = \frac{1}{2}\hat{B}$, $\hat{A}_2 = 50^\circ$ and $\hat{C}_1 = 100^\circ$.



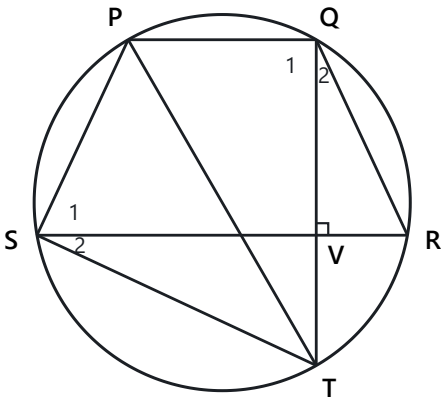
9.3.1 Calculate, with a reason, the size of $\hat{A}_1$. (3)

Asked: 2024

9.3.2 Hence prove that AD = DC. (5)

Asked: 2024

In the diagram, PQRS is a cyclic quadrilateral. T is a point on the circle such that QT ⊥ SR at V. PT and ST are drawn. $\hat{Q}_2 = 25^\circ$ and $\hat{R} = \hat{S}_1$.



9.4.1 Calculate, with reasons, the size of $\mathrm{Q\hat{T}S}$. (3)

Asked: 2025

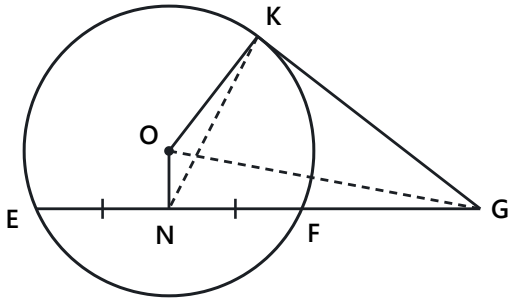
9.4.2 Prove that PQ ∥ SR. (4)

Asked: 2025, 2017 (MN ∥ TS)

9.4.3 Prove that PT is a diameter of the circle. (3)

Asked: 2025

In the diagram, O is the centre of the circle. N is the midpoint of chord EF. EF is produced to G and GK is a tangent to the circle at K. ON, OK, OG and KN are drawn.



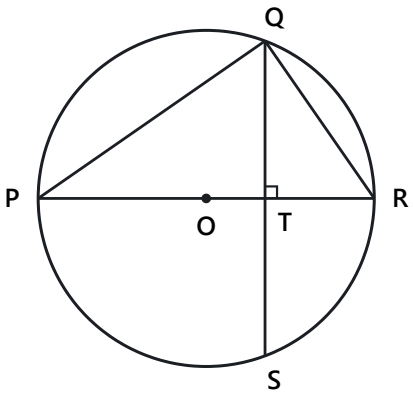
9.5.1 Prove that ONGK is a cyclic quadrilateral. (4)

Asked: 2017, 2020, 2022

9.5.2 Prove that $\mathrm{K\hat{O}G} = \mathrm{K\hat{N}G}$. (2)

Asked: 2022

In the diagram, PR is a diameter of the circle with centre O. Chord QS ⊥ PR at T. PQ and QR are drawn.



9.6.1 Give a reason why QT = TS. (1)

Asked: 2017, 2021

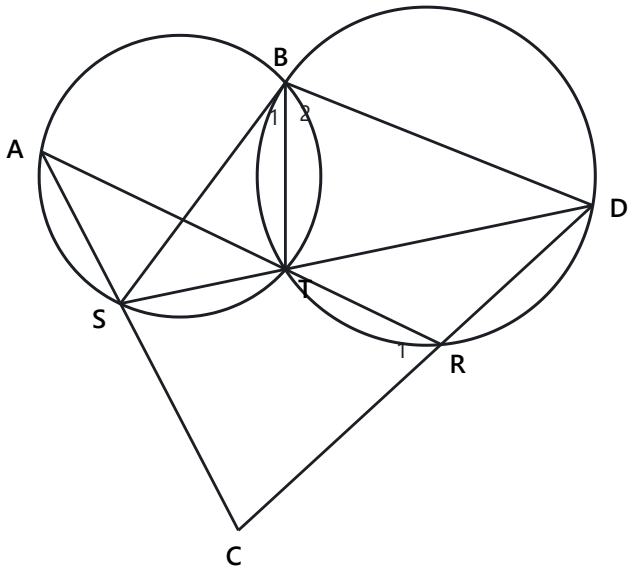
9.6.2 Give a reason why $\mathrm{P\hat{Q}R} = 90°$. (1)

Asked: 2016, 2018, 2020, 2021

9.6.3 Prove that PQ is a tangent to the circle passing through Q, T and R. (5)

Asked: 2017, 2021

Two circles intersect at B and T. A and S lie on the smaller circle; D and R lie on the larger circle. STD and ATR are straight lines. AS and DR are produced to meet at C. BS, BD and BT are drawn. Let $\mathrm{S\hat{A}T} = x$ and $\hat{R}_1 = y$.



9.7.1 Name, with a reason, another angle equal to x. (2)

Asked: 2018

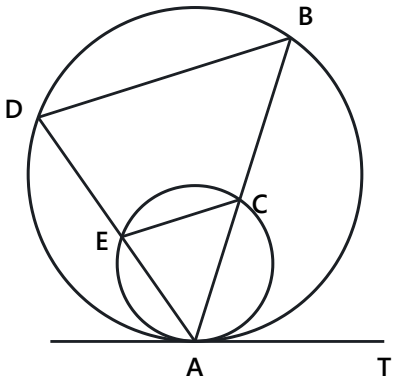
9.7.2 Name, with a reason, another angle equal to y. (2)

Asked: 2018

9.7.3 Prove that SCDB is a cyclic quadrilateral. (3)

Asked: 2018

Two circles touch internally at A. TA is a common tangent at A. Chord AB of the larger circle cuts the smaller circle at C and chord AD cuts it at E.



9.8.1 Prove that CE ∥ BD. (4)

Asked: 2024

9.8.2 Hence prove that $AC \cdot AD = AE \cdot AB$. (2)

Asked: 2024

9.9 In a diagram with a cyclic quadrilateral SBDC and $\hat{D}_2 = 30^\circ$ it is claimed that SD is a diameter. If you can prove that $\mathrm{S\hat{B}D} = 110^\circ$, explain why SD is NOT a diameter. (4)

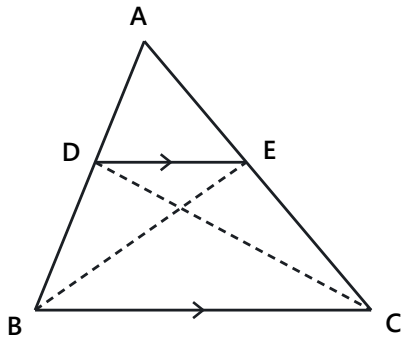
Asked: 2018

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QUESTION 10: EUCLIDEAN GEOMETRY: PROPORTIONALITY AND SIMILARITY [91 marks of variants]

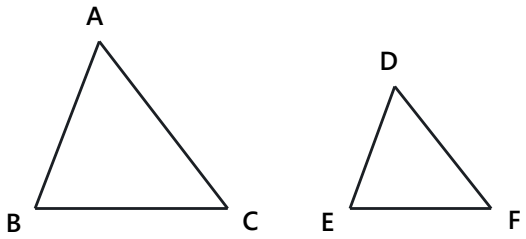
Every year: one numeric proportion question (line ∥ one side of a triangle, ratios, areas) and one similarity rider, usually inside a circle with a tangent (prove triangles similar, then a product like $DC^2 = DB \cdot DA$, then a length). The proportionality theorem proof came up in 2016, 2019 and 2024. 9–25 marks.

10.1 In △ABC, D lies on AB and E lies on AC such that DE ∥ BC. Prove the theorem that states that a line drawn parallel to one side of a triangle divides the other two sides proportionally, i.e. $\frac{AD}{DB} = \frac{AE}{EC}$. (6)



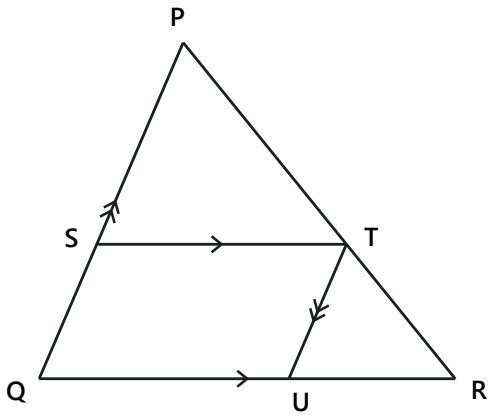
Asked: 2016, 2019, 2024

10.2 △ABC and △DEF have $\hat{A} = \hat{D}$, $\hat{B} = \hat{E}$ and $\hat{C} = \hat{F}$. Prove the theorem that states that if two triangles are equiangular, their corresponding sides are in proportion, i.e. $\frac{AB}{DE} = \frac{AC}{DF}$. (6)



Asked: Exam Guidelines (examinable proof)

In △PQR, S is on PQ and T is on PR such that ST ∥ QR. U is on QR such that TU ∥ PQ. PS : SQ = 3 : 2, PR = 30 and QR = 20.



10.3.1 Calculate the length of PT. (2)

Asked: 2017, 2018, 2023

10.3.2 Write down the ratio RU : UQ. (2)

Asked: 2017, 2025

10.3.3 Prove that △PST ||| △PQR and hence calculate ST. (3)

Asked: 2023 (prove △EDF ||| △EAB, then FC), 2023

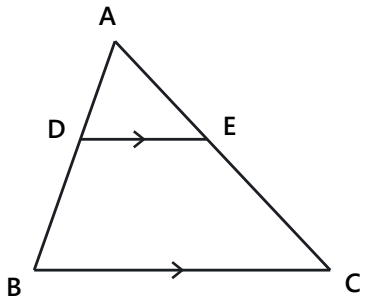
10.3.4 Calculate $\dfrac{\text{area } \triangle PST}{\text{area } \triangle PQR}$. (3)

Asked: 2024 (area ratio), 2025

10.3.5 Calculate $\dfrac{\text{area of parallelogram } SQUT}{\text{area } \triangle PQR}$. (4)

Asked: 2025 (area △GCF : area GFDA)

In △ABC, D lies on AB and E lies on AC with DE ∥ BC. AD : DB = 3 : 4, AE = $x + 1$ and EC = $2x - 2$.



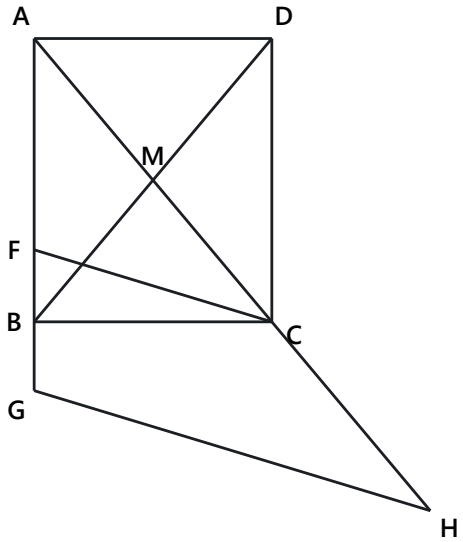
10.4.1 Calculate, with reasons, the value of x. (4)

Asked: 2020, 2022

10.4.2 If D and E were the MIDPOINTS of AB and AC, give the reason why DE ∥ BC and write down the relationship between DE and BC. (2)

Asked: 2021, 2022, 2024

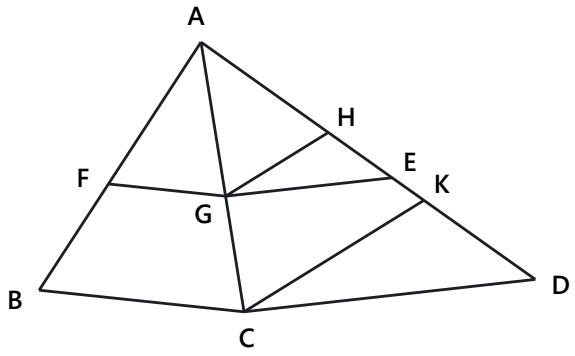
In △AGH, F lies on AG and C on AH with FC ∥ GH. AF = 12, FG = 8 and CH = 14. ABCD is a rectangle with B on AG; its diagonals AC and BD meet at M.



10.5.1 Calculate, with reasons, the length of DM. (5)

Asked: 2018

△ABC and △ACD share side AC. F is on AB, G is on AC, and H, E and K are on AD such that GH ∥ CK and GE ∥ CD. AF = $4x$, FB = $3x$, AG = $8y$ and GC = $6y$.



10.6.1 Prove that FG ∥ BC. (2)

Asked: 2017

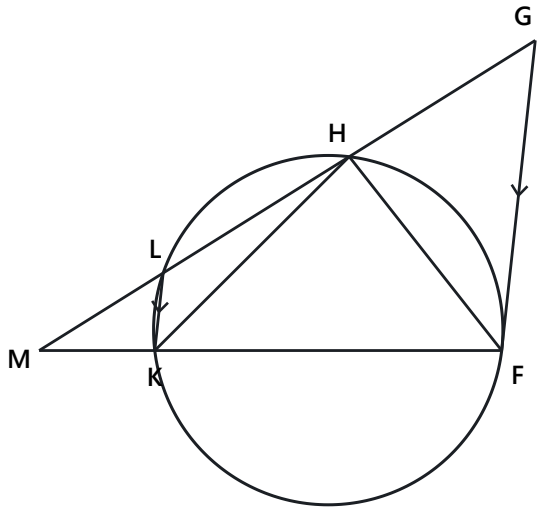
10.6.2 Prove that $\frac{AH}{HK} = \frac{AE}{ED}$. (3)

Asked: 2017

10.6.3 If AH = 8 and ED = 9, calculate EK. (4)

Asked: 2017

HLKF is a cyclic quadrilateral. Chords HL and FK are produced to meet at M. The line through F parallel to KL meets MH produced at G. MK = x, KF = $3x$, ML = y and LH = HG.



10.7.1 Prove that GH $= \frac{3}{2}y$. (3)

Asked: 2016

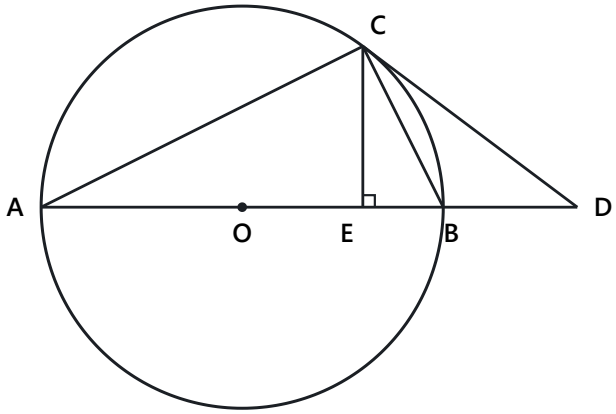
10.7.2 Prove that △MFH ||| △MGF. (5)

Asked: 2016

10.7.3 Show that $\frac{y}{x} = \sqrt{\frac{8}{5}}$. (3)

Asked: 2016

In the diagram, AB is a diameter of the circle with centre O. AB is produced to D and DC is a tangent to the circle at C. CE ⊥ AB. CA and CB are drawn. Let $\hat{\text{DCB}} = x$.



10.8.1 Name, with reasons, ONE other angle equal to x. (2)

Asked: 2020, 2021, 2022, 2024, 2025

10.8.2 Prove that △DCB ||| △DAC. (3)

Asked: 2020, 2021, 2022, 2025

10.8.3 Hence prove that $DC^2 = DB \cdot DA$. (2)

Asked: 2017, 2020, 2025

10.8.4 If DB = 4 units and the radius is 6 units, calculate DC. (3)

Asked: 2020, 2021

10.8.5 Prove that $\tan \hat{A} = \frac{1}{2}$. (3)

Asked: 2018 (AM/AB = $\sin^2$ x)

10.8.6 Prove that △CEB ||| △ACB and hence that $CB^2 = BE \cdot BA$. (4)

Asked: 2017 ($AD^2 = 3r^2$), 2018, 2024

10.8.7 Calculate CE. (4)

Asked: 2020 (DB), 2021 (LT)

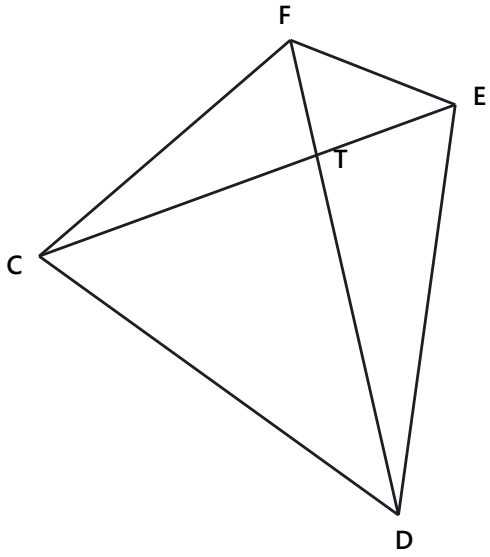
10.8.8 Prove that $DC^2 - DB^2 = DB \cdot BA$. (3)

Asked: 2022 (AC × SD = AR × TC), 2024 (GF^2 = ...), 2025 (PQ = QS)

10.8.9 Prove that $\frac{BE}{BA} = \sin^2 x$. (3)

Asked: 2018

The diagonals of quadrilateral EFCD intersect at T. EF = 6, DC = 15, ET = 5, TC = 10, FT = 4 and TD = 12,5.



10.9.1 Prove that $\triangle EFT \,|||\, \triangle DCT$. (4)

Asked: 2019

10.9.2 Prove that $D\hat{F}C = D\hat{E}C$. (3)

Asked: 2019

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